IB MATHEMATICS: APPLICATIONS & INTERPRETATION HL

Complete syllabus checklist

A clear topic-by-topic progress tracker covering all topics in the AI HL course.

○ LEARN	○ REVIEW	○ CONFIDENT

1 Number and algebra

37 lessons

NO.	LESSON			
1.1	Scientific notation	○	○	○
1.2.1	Arithmetic sequences and series	○	○	○
1.2.2	Sigma notation	○	○	○
1.3	Geometric sequences and series	○	○	○
1.4.1	Compound interest	○	○	○
1.4.2	Depreciation	○	○	○
1.4.3	Inflation	○	○	○
1.5.1	Laws of exponents	○	○	○
1.5.2	Introduction to logs	○	○	○
1.5.3	The number e	○	○	○
1.6.1	Rounding	○	○	○
1.6.2	Upper and lower bounds	○	○	○
1.6.3	Percentage error	○	○	○
1.7.1	Loans	○	○	○
1.7.2	Annuities	○	○	○
1.8.1	System of linear equations	○	○	○
1.8.2	Polynomial equations	○	○	○
1.9.1	Laws of logarithms	○	○	○
1.9.2	Change of Base	○	○	○
1.10	More Exponents	○	○	○
1.11	Sum of an infinite geometric series	○	○	○
1.12.1	Complex numbers introduction	○	○	○
1.12.2	Operations with complex numbers	○	○	○
1.12.3	Complex solutions to polynomial equations	○	○	○
1.13.1	Modulus-argument form	○	○	○
1.13.2	Euler form	○	○	○
1.13.3	Frequency and phase	○	○	○

1 Number and algebra (continued)

37 lessons

NO.	LESSON			
1.13.4	Geometric interpretation	○	○	○
1.14.1	Introduction to matrices	○	○	○
1.14.2	Algebra of matrices	○	○	○
1.14.3	Multiplication of matrices	○	○	○
1.14.4	Identity and zero matrices	○	○	○
1.14.5	Properties of matrices	○	○	○
1.14.6	Determinant and Inverse	○	○	○
1.14.7	System of linear equations	○	○	○
1.15.1	Eigenvalues and eigenvectors	○	○	○
1.15.2	Powers of matrices	○	○	○

2 Functions

29 lessons

NO.	LESSON			
2.1.1	Gradient of a line	O	O	O
2.1.2	Equation of a straight line	O	O	O
2.1.3	Parallel and perpendicular lines	O	O	O
2.2.1	Functions introduction	O	O	O
2.2.2	Evaluating functions	O	O	O
2.2.3	Inverse functions	O	O	O
2.3	Sketching functions	O	O	O
2.4.1	Features of graphs	O	O	O
2.4.2	Asymptotes	O	O	O
2.5.1	Linear modelling	O	O	O
2.5.2	Quadratic functions	O	O	O
2.5.3	Quadratic modelling	O	O	O
2.5.4	Exponential modelling	O	O	O
2.5.5	Direct and inverse variation	O	O	O
2.5.6	Cubic modelling	O	O	O
2.5.7	Sinusoidal graphs	O	O	O
2.5.8	Creating a sinusoidal model	O	O	O
2.6	Finding parameters of a model	O	O	O
2.7.1	Composite functions	O	O	O
2.7.2	Finding the inverse functions	O	O	O
2.8.1	Transformations	O	O	O
2.8.2	Composite transformations	O	O	O
2.9.1	Half life	O	O	O
2.9.2	Natural logarithmic modelling	O	O	O
2.9.3	Sinusoidal modelling extension	O	O	O
2.9.4	Logistic modelling	O	O	O
2.9.5	Piecewise models	O	O	O
2.10.1	Scaling using logs	O	O	O
2.10.2	log log graphs	O	O	O

3 Geometry and trigonometry

57 lessons

NO.	LESSON			
3.1.1	Midpoint of a line segment	○	○	○
3.1.2	Distance between two points	○	○	○
3.1.3	Volume and surface area	○	○	○
3.1.4	Trigonometry in 3D	○	○	○
3.2.1	Pythagoras’ theorem	○	○	○
3.2.2	SOH CAH TOA	○	○	○
3.2.3	Sine rule	○	○	○
3.2.4	Cosine rule	○	○	○
3.2.5	Area of a triangle	○	○	○
3.3.1	Bearings	○	○	○
3.3.2	Angles of elevation and depression	○	○	○
3.4	Arcs and sectors	○	○	○
3.5	Perpendicular bisectors	○	○	○
3.6.1	Voronoi diagram intro	○	○	○
3.6.2	Voronoi examples	○	○	○
3.7.1	Introduction ro radians	○	○	○
3.7.2	Arcs and sectors	○	○	○
3.8.1	Mr. Flynn's exact triangles	○	○	○
3.8.2	Unit circle	○	○	○
3.8.3	The CAST diagram	○	○	○
3.8.4	Graphs of sin, cos and tan	○	○	○
3.8.5	Sine ule ambiguous case	○	○	○
3.8.6	Trigonometric identities	○	○	○
3.8.7	Solving trigonometric equations	○	○	○
3.9.1	Matrix transformations (intro and stretches)	○	○	○
3.9.2	Translations	○	○	○
3.9.3	Enlargements	○	○	○
3.9.4	Rotations	○	○	○
3.9.5	Reflections	○	○	○

3 Geometry and trigonometry (continued)

57 lessons

NO.	LESSON			
3.9.6	Composite transformations	O	O	O
3.9.7	Geometric interpretation of determinant	O	O	O
3.10.1	Introduction to vectors	O	O	O
3.10.2	Operations with vectors	O	O	O
3.10.3	Magnitude and unit vectrs	O	O	O
3.11	Vector equation of a line	O	O	O
3.12.1	Vectors and kinematics	O	O	O
3.12.2	Motion with variable velocity	O	O	O
3.13.1	Scalar product and angle between vectors	O	O	O
3.13.2	Angle between two lines	O	O	O
3.13.3	Vector product	O	O	O
3.13.4	Areas using the vector product	O	O	O
3.13.5	Components of vectors	O	O	O
3.14.1	Graph theory introduction	O	O	O
3.15.1	Adjaceny matrices	O	O	O
3.15.2	Weighted adjacency tables	O	O	O
3.15.3	Walks	O	O	O
3.16.1	Graph routes	O	O	O
3.16.2	Kruskal's algorithm	O	O	O
3.16.3	Prim's algorithm	O	O	O
3.16.4	Prim's algorithm with adjacency table	O	O	O
3.16.5	Eulerian graphs	O	O	O
3.16.6	Chinese postman problem	O	O	O
3.16.7	Hamiltonian graphs	O	O	O
3.16.8	Travelling salesman problem	O	O	O
3.16.9	TSP practical problems	O	O	O
3.16.10	Nearest neighbour algorithm	O	O	O
3.16.11	Deleted vertex algorithm	O	O	O

4 Statistics and probability

53 lessons

NO.	LESSON			
4.1.1	Discrete vs continuous data	O	O	O
4.1.2	Sampling techniques	O	O	O
4.2.1	Histograms and cumulative frequency curves	O	O	O
4.2.2	Box and whisker diagrams	O	O	O
4.2.3	Interpreting box plots	O	O	O
4.2.4	Outliers	O	O	O
4.3.1	Averages and spread	O	O	O
4.3.2	Frequency tables	O	O	O
4.3.3	Constant changes in data	O	O	O
4.3.4	Variance and standard deviation by hand	O	O	O
4.4	Linear regression and correlation	O	O	O
4.5	Probability formulae	O	O	O
4.6.1	Venn diagrams	O	O	O
4.6.2	Tree diagrams	O	O	O
4.7	Discrete random variables	O	O	O
4.8	Binomial distribution	O	O	O
4.9	Normal distribution	O	O	O
4.10	Spearman’s rank correlation coefficient	O	O	O
4.11.1	Chi squared intro	O	O	O
4.11.2	Chi squared test for independence	O	O	O
4.11.3	Chi squared goodness of fit	O	O	O
4.11.4	T-test	O	O	O
4.12.1	Surveys and questionnaires	O	O	O
4.12.2	Reliability vs validity	O	O	O
4.12.3	Chi squared (categorising data)	O	O	O
4.12.4	Chi squared (degrees of freedom)	O	O	O
4.13.1	Regression with non-linear functions	O	O	O

4 Statistics and probability (continued)

53 lessons

NO.	LESSON			
4.13.2	Sum of square residuals	○	○	○
4.13.3	R squared	○	○	○
4.14.1	Linear transformations of X	○	○	○
4.14.2	Linear combinations of n random variables.	○	○	○
4.14.3	Unbiased estimates for mean and variance	○	○	○
4.14.4	Unbiased estimates example questions	○	○	○
4.15.1	Linear combination of n independent normal random variables	○	○	○
4.15.2	Normal distribution sampling	○	○	○
4.15.3	Central limit theorem	○	○	○
4.16.1	Confidence intervals (known variance)	○	○	○
4.16.2	Confidence intervals (unknown variance)	○	○	○
4.17	Poisson distribution	○	○	○
4.18.1	Z-test one tailed	○	○	○
4.18.2	Z-test two tailed	○	○	○
4.18.3	T-test	○	○	○
4.18.4	Matched pairs	○	○	○
4.18.5	Testing for populations proportion	○	○	○
4.18.6	Test for mean using Poisson distribution	○	○	○
4.18.7	Testing for correllation	○	○	○
4.18.8	Type I and Type II errors	○	○	○
4.18.9	Normal errors	○	○	○
4.18.10	Poisson errors	○	○	○
4.18.11	Binomial errors	○	○	○
4.19.1	Markov chains	○	○	○
4.19.2	Transition matrices	○	○	○
4.19.3	Steady state and long term probabilities	○	○	○

5 Calculus

50 lessons

NO.	LESSON			
5.1.1	Introduction to differentiation	○	○	○
5.1.2	Introduction to limits	○	○	○
5.2	Increasing and decreasing functions	○	○	○
5.3.1	Differentiation power rule	○	○	○
5.3.2	Derivative at a point	○	○	○
5.4	Tangents and normals	○	○	○
5.5.1	Integration (power rule)	○	○	○
5.5.2	Finding f(x)	○	○	○
5.5.3	Area under the curve	○	○	○
5.6	Stationary (max/min) points	○	○	○
5.7	Optimisation	○	○	○
5.8	Trapezoidal rule	○	○	○
5.9.1	Differentiating trigonometric, e^x and ln x	○	○	○
5.9.2	Differentiating x^n, where n is rational	○	○	○
5.9.3	Chain Rule	○	○	○
5.9.4	Product Rule	○	○	○
5.9.5	Quotient rule	○	○	○
5.9.6	Related rates of change	○	○	○
5.10.1	Second derivative	○	○	○
5.10.2	Relationship between graphs and derivatives	○	○	○
5.10.3	Classifying stationary points	○	○	○
5.10.4	Points of inflexion	○	○	○
5.11.1	Integration of x^n, where n is rational	○	○	○
5.11.2	Integrating trig	○	○	○
5.11.3	Integrating e^x	○	○	○

5 Calculus (continued)

50 lessons

NO.	LESSON			
5.11.4	Inverse chain rule	○	○	○
5.11.5	Integration by substitution	○	○	○
5.12.1	Area between curve and x axis	○	○	○
5.12.2	Area between curve and y axis	○	○	○
5.12.3	Volume of revolution about x axis	○	○	○
5.12.4	Volume of revolution about y axis	○	○	○
5.13.1	Kinematics	○	○	○
5.13.2	Kinematics 2	○	○	○
5.14.1	Setting up differential equation	○	○	○
5.14.2	Solving by separation of variables	○	○	○
5.15	Slope field and their diagrams	○	○	○
5.16.1	Euler's method	○	○	○
5.16.2	Euler's method example	○	○	○
5.16.3	Euler's method for coupled systems	○	○	○
5.17.1	Phase portraits intro	○	○	○
5.17.2	Solutions of coupled differential equations (p1)	○	○	○
5.17.3	Solutions of coupled differential equations (p2 and saddle)	○	○	○
5.17.4	Sketching phase portraits introduction	○	○	○
5.17.5	Sketching phase portraits (source)	○	○	○
5.17.6	Sketching phase portraits (sink)	○	○	○
5.17.7	Sketching phase portraits (spiral sink)	○	○	○
5.17.8	Sketching phase portraits (spiral source)	○	○	○
5.17.9	Sketching phase portraits (centre)	○	○	○
5.18.1	Second order differential equations (Euler's method)	○	○	○
5.18.2	Second order differetnial equations (exact solutions)	○	○	○